

Inputs

from Stages 2 and 3

- $G = (V, E)$ — method-level function call graph, from Androguard
- E — 320-dimensional code2vec embedding per function
- S — backward slices around suspicious API call sites

Hyper-parameters

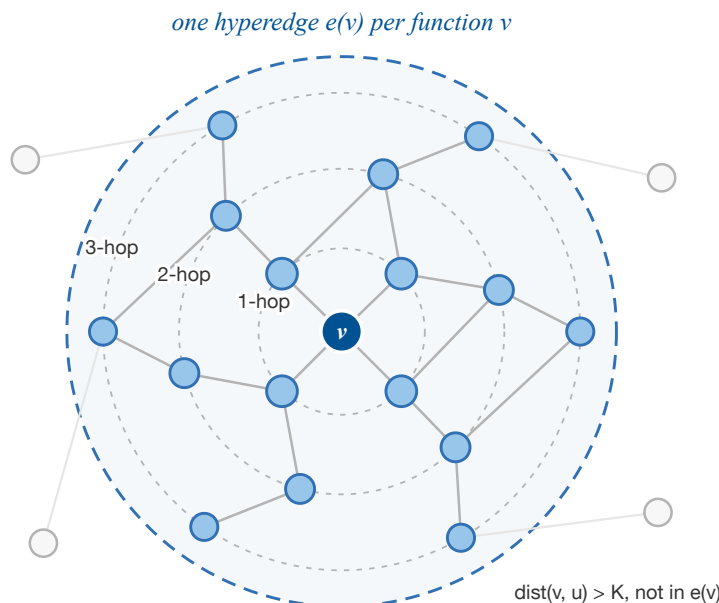
- K call-graph hop radius, 3
- k semantic neighbours, 5
- τ similarity floor, 0.80
- L_p maximum path in-fill length
- c_{max} hyperedge cardinality cap, 64

Every construction emits the same triple — (e, w, t) — a node set, a scalar weight and a one-byte type tag, so the three views stay separable downstream.

4 Hyperedge generation Structure, semantics and security-critical data flow over one vertex set

4A K-hop topological

hyperedge set H_K , type tag **topo**



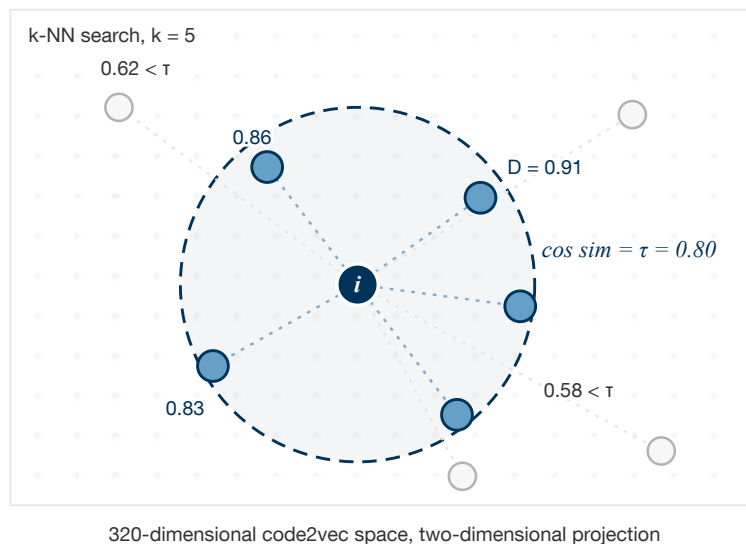
$$e(v) = \{ u \in V : \text{dist}_G(v, u) \leq K \}, \quad 2 \leq |e| \leq c_{\max}$$

One hyperedge per function: its **K-order callers and callees** collapsed into a single higher-order relation. Uniform weight $w = 1$, because topology asserts adjacency, not importance.

● anchor v ● member, $\text{dist} \leq K$ ● excluded

4B Semantic k-nearest neighbour

hyperedge set H_{SEM} , type tag **sem**



$$e(i) = \{ I_{ij} : D_{ij} \geq \tau \}, \quad k = 5, \quad \tau \approx 0.80$$

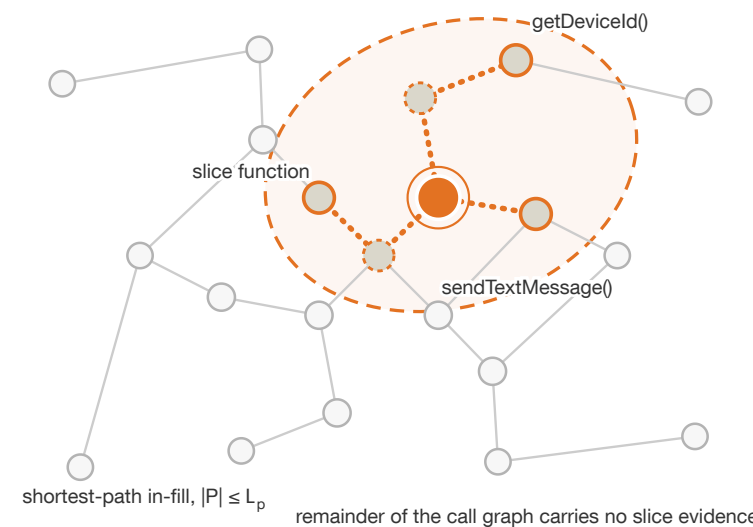
A k-nearest neighbour query over the code2vec embeddings, gated by the similarity floor τ . This groups **functions that behave alike but never call one another**, the signal for code reuse, repackaging and polymorphic variants that a call graph cannot express.

● query function ● neighbour, $\text{sim} \geq \tau$ ● candidate dropped

4C Slice-aware suspicious API

hyperedge set H_{API} , type tag **sus**

one hyperedge per backward slice $s \in S$



$$w = 1 + 0.2 \times (\# \text{transmit}) + 0.1 \times (\# \text{access})$$

Every function a backward slice touches forms one hyperedge; shortest paths between slice members of length at most L_p are folded in so the exfiltration chain stays connected. The weight **prices transmission above access**: data leaving the device outranks data merely being read.

$$2 \text{ transmit and } 3 \text{ access calls give } w = 1 + 0.4 + 0.3 = \mathbf{1.7}$$

● suspicious call site ● function in slice ● path in-fill